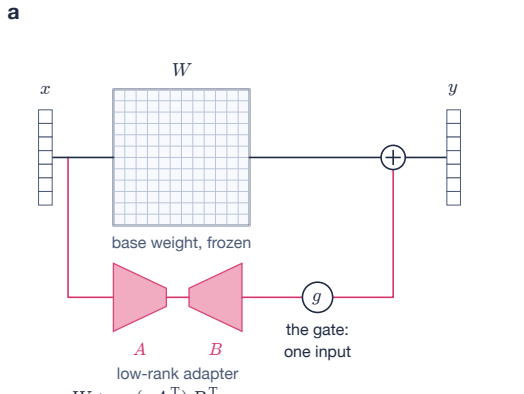
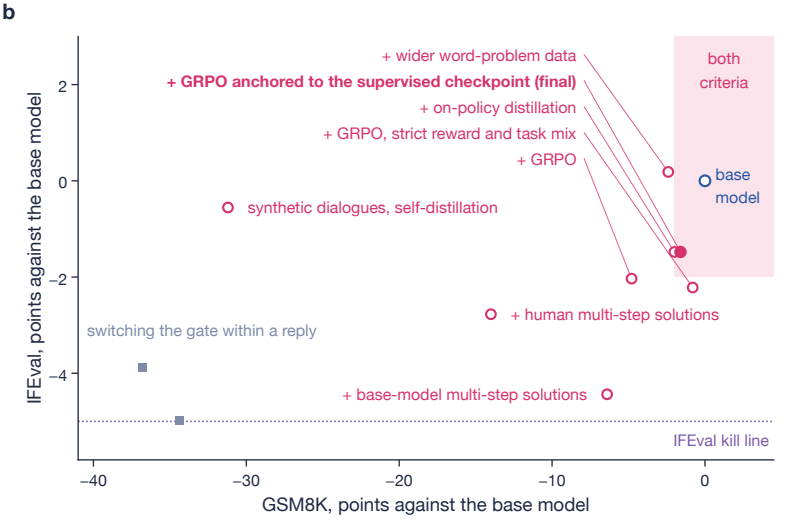
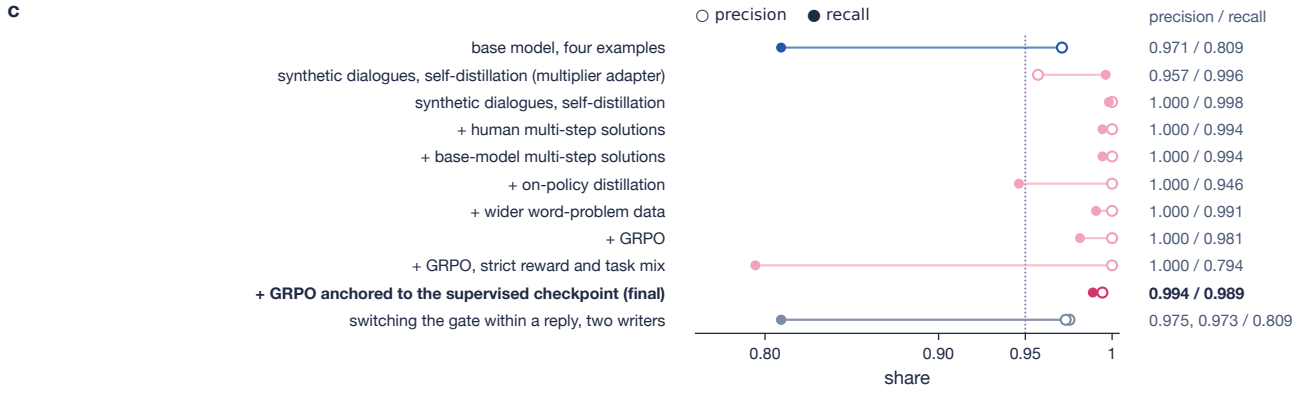

a
W
x
y
base weight, frozen
g
A
B
the gate:
one input
low-rank adapter
$y = xW + g\,s\,(xA^{\mathrm{T}})\,B^{\mathrm{T}}$
$g = 0$: the base model, Qwen3.5-0.8B
$g = 1$: the sheet machine
each bit for bit, on 2,500 chat forward passes and 2,688 table views


b
+ wider word-problem data
+ GRPO anchored to the supervised checkpoint (final)
+ on-policy distillation
+ GRPO, strict reward and task mix
+ GRPO
both
criteria
base
model
synthetic dialogues, self-distillation
+ human multi-step solutions
switching the gate within a reply
+ base-model multi-step solutions
IFEval kill line
IFEval, points against the base model
GSM8K, points against the base model
2
0
−2
−4
−40
−30
−20
−10
0


c
precision
recall
precision / recall
base model, four examples
0.971 / 0.809
synthetic dialogues, self-distillation (multiplier adapter)
0.957 / 0.996
synthetic dialogues, self-distillation
1.000 / 0.998
+ human multi-step solutions
1.000 / 0.994
+ base-model multi-step solutions
1.000 / 0.994
+ on-policy distillation
1.000 / 0.946
+ wider word-problem data
1.000 / 0.991
+ GRPO
1.000 / 0.981
+ GRPO, strict reward and task mix
1.000 / 0.794
+ GRPO anchored to the supervised checkpoint (final)
0.994 / 0.989
switching the gate within a reply, two writers
0.975, 0.973 / 0.809
0.80
0.90
0.95
1
share